Que1. A user opens your portfolio from a link. Describe step by step what happens between the browser and the server, and draw a simple client–server diagram (hand-drawn and photographed is fine). Say where the app logic and the database sit.

---→ when user click on link it open in browser here browser act as client

And send request to server.

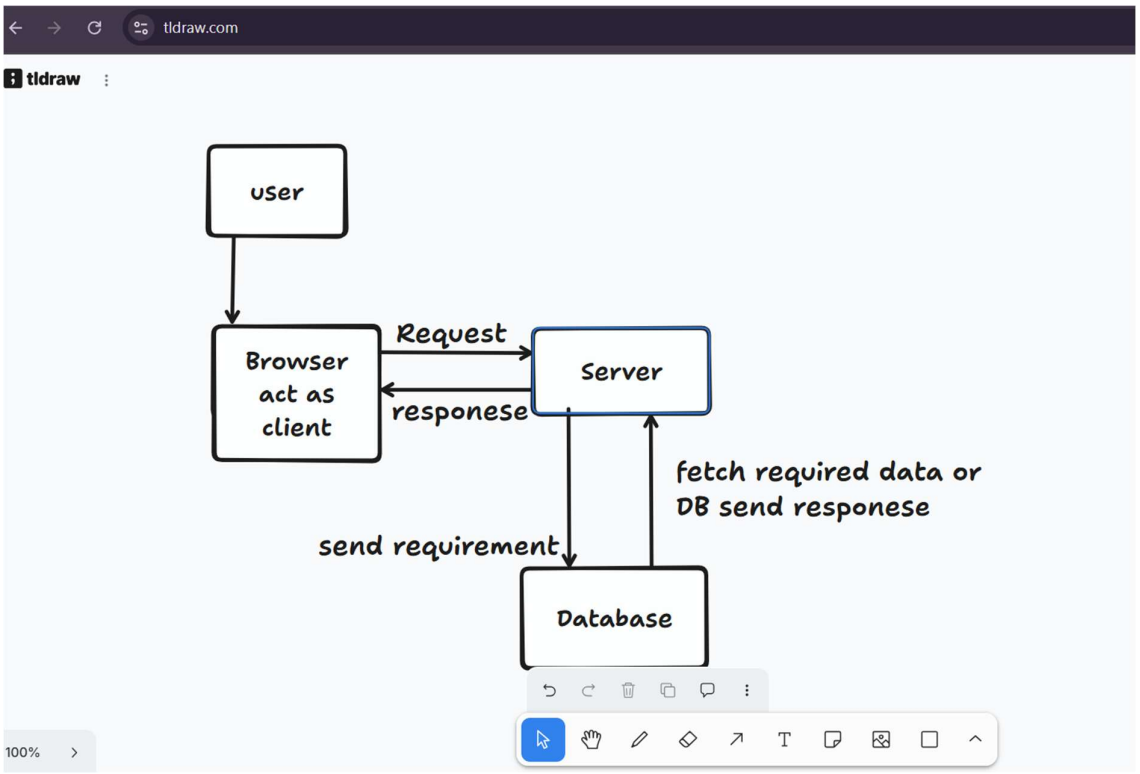
Who is the server?

The computer or software who host my website.

Then server accept request and according to request fetch information from database and give as response to client.

The app logic and database sit at server side.we write our app logic in javascript and

And every browser have a by default interprete of javaScript.so app logic and database sit at server side.



Que 2. Your contact form sends a visitor's message to a server. Should it use GET or POST, and why? What happens if method is left out? Name the status code the server returns on success and the family of codes that means the server failed.

--→Visitor's message send to server means here we use POST method

Because we don't have to retrive information instead of we have to submit contact information of user to the server.so here we use POST method.

If method is left out then form has default method GET because in form we have to collect information from user .if POST method is left out then GET method executed and code 200 is shown that is successful.

200 OK:successful

201 created:created

5xx family of code that means server failed

500 :internal server error

502 :bad gateway

503 :server unavailable

504 :gateway timeout

Que 3. Name the three types of CSS and say where each is written. If all three colour the same paragraph differently, which wins? Which type would you use for a style shared by four pages, and why?

--→There are Three type of css

1.Inline css: write directly inside html element tag.

For example:<body>

<h1 style="color:Red ;Font-size:36px">Hello World</h1>

</body>

2.External css:

External css means it is separate .css file and this .css file link with html file

For example:

Index.html----→

<html>

<head>

<link rel=”stylesheet” type=”text/css” href=”style.css”>

</head>

</html>

Style.css-→

Body{

Background-color:red;

}

P{

Color:green

}

3.Internal css

Internal css means inside <style> block and inside <head> block of html and it affects the entire html page.

For example:

Oue 4. A page has Hello . Rules exist for *, p, each with a different .note and #intro, color. Which applies, and what is the full order of precedence from strongest to weakest?

---→*{

Color:black;

}

P{

color:blue;

}

.note{

Color:green;

}

#intro{

Color:red;

}

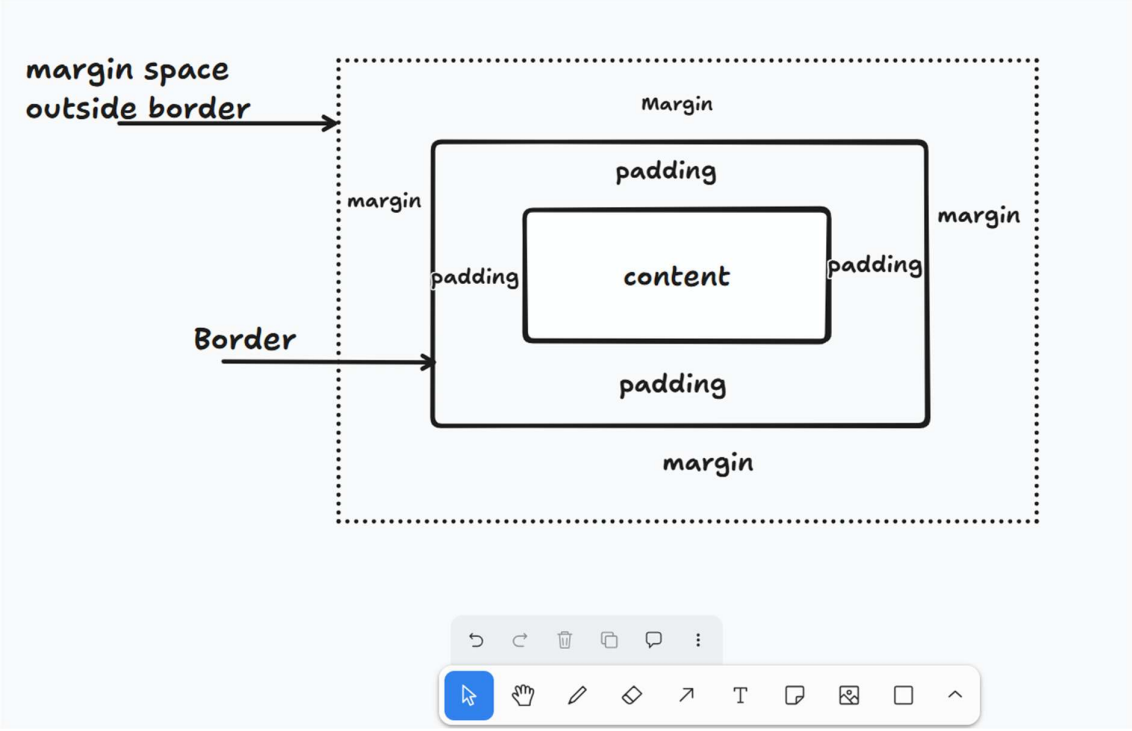
The color for Hello is red because id selector has high preference

The preference is as follows:

Id Selector>class Selector>element Selector>Universal selector.

Que 5. Explain padding and margin with a small drawing of a showing content, padding, border and margin. What does box-sizing: border-box change?

→the box-sizing: border-box property includes padding and border within the specified width and height, making element sizing easier to manage.



Que. 6. Where does JavaScript code run, and what is the JS engine in a browser? Why is no compiler needed to run JavaScript? Name two things JavaScript can do on a web page that HTML and CSS alone cannot.

-→javascript code run inside web browser or node.js

In a browser javascript can run on computer .java script engine is a software that reads,processes and execute javascript code.

Javascript does not require seprate compiler it uses just in time compilation or we called interpretation.

Where javascrpt can do that html and css alone cannot:

1.validate form:

Check whether the user enter valid email address or required information.

2.respond to user action dynamically:

When user clicks a button javascript can change text,calculate result,or show and hide content.

Que 7. JavaScript has no type safety. Explain what happens when you write then a = "hello";. Name the three data types you have learned so far.

-→a=hello;

A=5; the same variable now hold a number

Java script allows this without requiring us to declare the variable type.

Javascript datatypes:

1.Number 2.String 3.Boolean

Que 8.Explain the four values of the CSS position property (static and relative, absolute, fixed). When would you use top, left, right and bottom?

-→ css position property controls element placement.

Static follows normal flow

Relative moves from the original position

Absolute position an element relative to qualifying ancesotor and fixed generally stays relative to the viewport.

The top,right,left,bottom properties control offset for positioned elements.

Que 9. A container has display: flex and three boxes with flex values of 1, 2 and 1. How is the width shared between the boxes? What do gap and justify-content: center do?

- ➔ Box 1=25%

Box 2=50%

Box 3=25%

Gap:

Gap adds space between flex items.

```
.container{
Display:flex;
Gap:20px;
}
```

Justify-content:

Positions the flex items together at the center of the container along the main axis.

```
.container{
Display:flex;
Justify-center:center;
}
```